

Chemotaxis, collective dynamics, and diffusive relays in neutrophil swarming

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Neutrophils, the most abundant immune cells, swarm collectively around infections and injuries. Recent experiments show that swarms are organized not by static gradients but by self-propagating waves of chemoattractant: stimulated cells relay the signal to their neighbours, while a negative feedback loop stops the relay and limits its range [1]. A single cell, by contrast, makes net progress toward the source only over a narrow window of wave speeds [2]. The classical Keller–Segel model misses this, predicting net motion away from the source at all speeds.

We model this in two parts. First, we treat the relay as an excitable reaction-diffusion system and compare several inhibition mechanisms that could stop the wave and set the swarm size; the resulting fronts are pushed waves. Second, we couple cell motility to the cue field through the cell polarization, an internal degree of freedom that sets the direction of motion and relaxes on its own timescale. This lag breaks the symmetry of a passing pulse and gives a net drift, reproducing the experimental observations of [2]. Combining the two, where cells both make the wave and move in it, gives a self-consistent picture of neutrophil swarming.

[1] E. Strickland, D. Pan, C. Godfrey, J. S. Kim, A. Hopke, W. Ji, M. De-grange, B. Villavicencio, M. K. Mansour, C. S. Zerbe, D. Irimia, A. Amir, and O. D. Weiner, *Self-extinguishing relay waves enable homeostatic control of human neutrophil swarming*, *Developmental Cell* **59**, 2659, (2024).

[2] M. Ishida, M. Uwamichi, A. Nakajima, and S. Sawai, *Traveling-wave chemotaxis of neutrophil-like HL-60 cells*, *Molecular Biology of the Cell* **36**, ar17, (2025).